

Unit 3: Monte Carlo Methods in Inference

Chapter 7 in “Statistical Computing with R”

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February 23, 2026

1. Introduction
2. R Programming DOs and DON'Ts
3. Estimation
4. (Aside) Rant About Statistics
5. Hypothesis Testing

Introduction

Introduction

- Monte Carlo methods are helpful for estimation.
- You will likely employ these types of techniques if you do a thesis within statistics.
- We'll use it to estimate the following:
 - Mean
 - Standard Error
 - MSE (Mean squared error)
 - Alternatives to MSE
 - Confidence Levels
- We can also use Monte Carlo methods for hypothesis testing and generating p -values.

The following questions motivate the discussion ahead:

- What is the goal of statistics?
- Why do we perform hypothesis tests?
- What is actually going on when we do a hypothesis test?
- Do we actually make reasonable assumptions for the null hypothesis?
- Why do we have the threshold $\alpha = 0.05$?
- Why do the mathematicians mock statisticians?
- Can we actually start changing statistics education?

These questions are addressed in a dedicated aside later in the unit.

R Programming DOs and DON'Ts

R Programming DOs and DON'Ts

- It may seem late to introduce these practices, but there is a pedagogical reason for the delay.
- Introducing coding best practices earlier would require balancing several competing demands simultaneously:
 - Teaching you how to code.
 - Teaching you best practices.
 - Teaching you how to write tests.
- There is a genuine pedagogical tension here: efficient code tends to be harder to read and understand.
- Learning through suboptimal methods has real costs, as I discovered firsthand when transitioning to professional R development.
- Now that the course covers more complex computational structures, it is the appropriate moment to establish better habits.

R Programming DOs and DON'Ts

- Absolute optimisation is not the goal here; the aim is to eliminate obviously inefficient habits.
- Naturally, when coding, you will think of the inefficient solution first, which is natural.
- However, you should recognise that a more efficient solution exists and practice writing them.
- Efficiency becomes especially consequential in large-scale projects.
- This is obviously not comprehensive. There is a more detailed guide here. (Click text).
- We will not deduct marks in examinations for inefficient code, UNLESS you do the number one sin of growing R objects.
- For the project, there will be marks deducted if I comment on the inefficiencies and you do not fix them later.

Appending Entries Is Inefficient!

Inefficient For Loop and Appending

```
n = 100; x1 <- rnorm(n);  
# NEVER DO  
est <- c()  
for(i in 1:n){  
  val <- x1[i]^2  
  est <- c(est, val)  
}  
# INSTEAD  
est2 <- x1^2
```

Ask yourself: did you need to write a for-loop for your operation?

...I Saw This in a Textbook Once

Repeating

```
# NEVER DO
v1 <- c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
# INSTEAD (either or)
v2 <- rep(0, 10)
v3 <- numeric(10) # 0 case only
```

Easy Integer List

```
# NEVER DO
v1 <- c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
# INSTEAD
v2 <- 1:10
```

Matrices Are Efficient!

You will see in the examples for MC estimation that I will be using a lot of matrices!
Some elementary functions below:

Basic Matrix Use

```
n <- 10000  
m <- matrix(rnorm(n*n), nrow = n)  
rowSums(m)  
colSums(m)  
rowMeans(m)  
colMeans(m)
```

How to Test Efficiency?

system.time()

```
> n = 100; x1 <- rnorm(n);  
> # NEVER DO  
> system.time({  
+   est <- c()  
+   for(i in 1:n){  
+     val <- x1[i]^2  
+     est <- c(est, val)  
+   }  
+ })  
      user  system elapsed  
0.002    0.000    0.002
```

How to Test Efficiency?

```
system.time()
```

```
> # INSTEAD  
> system.time({  
+   est2 <- x1^2  
+ })  
      user  system elapsed  
       0      0        0
```

Monte Carlo Estimation

General Idea for MC Estimation

- Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution X .
- An estimator $\hat{\theta}$ for a parameter θ is a function of the random sample.
- Random variates from the sampling distribution of X can be generated by repeatedly drawing independent random samples $\mathbf{x}^{(j)}$ for $j = 1, 2, \dots, m$ where m represents the Monte Carlo sample size.
- We can compute $\hat{\theta}$ for each sample j .

Estimating the Mean

Example

Suppose that X_1, X_2 are iid from a standard Gaussian distribution.

- (a) Find the exact value of $\mathbb{E}[|X_1 - X_2|]$.
- (b) Derive a Monte Carlo estimator of $\mathbb{E}[|X_1 - X_2|]$.
- (c) Use part (b) to find an approximate value of $\mathbb{E}[|X_1 - X_2|]$.

Solution. Consider $Y = X_1 - X_2$. Then,

$$\mathbb{E}[Y] = \mathbb{E}[X_1 - X_2] = \mathbb{E}[X_1] - \mathbb{E}[X_2] = 0 - 0 = 0$$

$$\mathbb{V}[Y] = \mathbb{V}[X_1 - X_2] = \mathbb{V}[X_1] + \mathbb{V}[X_2] - 2\text{Cov}(X_1, X_2) = 1 + 1 - 0 = 2$$

So $Y \sim N(\mu = 0, \sigma^2 = 2)$.

Estimating the Mean

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\} = \frac{1}{\sqrt{4\pi}} \exp \left\{ -\frac{y^2}{4} \right\} = \frac{1}{2\sqrt{\pi}} \exp \left\{ -\frac{y^2}{4} \right\}$$

Then,

$$\begin{aligned} \mathbb{E}[|Y|] &= \int_{-\infty}^{\infty} |y| \frac{1}{2\sqrt{\pi}} \exp \left\{ -\frac{y^2}{4} \right\} dy \\ &= \int_0^{\infty} y \frac{1}{2\sqrt{\pi}} \exp \left\{ -\frac{y^2}{4} \right\} dy + \int_{-\infty}^0 -y \frac{1}{2\sqrt{\pi}} \exp \left\{ -\frac{y^2}{4} \right\} dy \\ &= 2 \int_0^{\infty} y \frac{1}{2\sqrt{\pi}} \exp \left\{ -\frac{y^2}{4} \right\} dy \end{aligned}$$

Let $u = y^2/4$. Then, $du = \frac{y}{2} dy$ and thus:

Estimating the Mean

$$\mathbb{E}[|Y|] = 2 \int_0^{\infty} \frac{1}{\sqrt{\pi}} e^{-u} du = \frac{2}{\sqrt{\pi}} \left[-e^{-u} \right]_0^{\infty} = \frac{2}{\sqrt{\pi}} \left[\lim_{u \rightarrow \infty} -e^{-u} + e^0 \right] = \frac{2}{\sqrt{\pi}}$$

To derive the Monte Carlo estimator, we generate random samples $\mathbf{x}^{(j)} = (x_1^{(j)}, x_2^{(j)})$ where $j = 1, 2, \dots, n$ and n represents the number of total replicates. We compute the mean of the replicates:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n |x_1^{(j)} - x_2^{(j)}|$$

See the accompanying R code.

Critiquing R-code

R Code from Textbook

```
m <- 10000
g <- numeric(m)
for (i in 1:m) {
  x <- rnorm(2)
  g[i] <- abs(x[1] - x[2])
}
val2 <- mean(g)
```

Some glaring issues:

- Unnecessary for-loop
- `rnorm()` is called m times unnecessarily; a single vectorised call at the outset would suffice.

Estimating Standard Error

Standard Error

Standard error of the mean $\bar{X}$ of a sample of size n is $\sqrt{\widehat{\mathbb{V}}(X)/n}$.

Estimators for Standard Error

An intuitive choice to estimate the variance is:

$$\widehat{\mathbb{V}}(X) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

But the **unbiased** estimator is:

$$\widehat{\mathbb{V}}(X) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Estimating Standard Error

Example

In the previous example, find the estimate of the standard error for $|X_1 - X_2|$ using two methods. Compare it to the true standard error.

Solution. The derivation of the theoretical standard error proceeds as follows. Consult the accompanying R code.

$$\mathbb{V}[|Y|] = \mathbb{E}[|Y|^2] - \mathbb{E}[|Y|]^2$$

Recall that $|Y|^2 = Y^2$ and thus $\mathbb{E}[|Y|^2] = \mathbb{E}[Y^2]$. We can easily find this from:

$$\mathbb{V}[Y] = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \Rightarrow \mathbb{E}[Y^2] = \mathbb{V}[Y] + \mathbb{E}[Y]^2 = 2 + 0$$

Thus,

$$\mathbb{V}[|Y|] = 2 - \left(\frac{2}{\sqrt{\pi}}\right)^2 = 2 - \frac{4}{\pi} \Rightarrow SE(|Y|) = \sqrt{\frac{\mathbb{V}[|Y|]}{n}} = \sqrt{\frac{2(1 - \frac{2}{\pi})}{n}}$$

Mean Squared Error (MSE)

$$MSE(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \mathbb{V}[\hat{\theta}] + Bias(\hat{\theta})^2$$

Based on this formula, the Monte Carlo estimate can be found using the following:

1. Generate m random samples $x^{(1)}, x^{(2)}, \dots, x^{(m)}$ from the distribution of interest X , where $x^{(i)}$ are of size n for $i = 1, 2, \dots, m$.
2. Compute the estimator: $\hat{\theta}^{(j)} = \hat{\theta}(x_1^{(j)}, x_2^{(j)}, \dots, x_n^{(j)})$ for $j = 1, 2, \dots, m$.
3. Compute: $\widehat{MSE}(\hat{\theta}) = \frac{1}{m} \sum_{j=1}^m (\hat{\theta}^{(j)} - \theta)^2$.

Warning: do not confuse m and n in this case!!

Estimating MSE

Example

Suppose $X_1, X_2, \dots, X_n$ are iid $N(\theta, \theta^2)$. We want to estimate θ . We have two possible candidates:

$$\hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{\theta}_2 = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Compute the MSE of these two estimators and determine which one is “better”.

Solution. Consult the R codes. The second estimator should be “better”.

Estimating Alternatives to MSE

- Why do we use the MSE? Karlin (1958) noted advantages of using MSE when comparing estimators:
 - For an unbiased estimator $\hat{\theta}$ of θ , the MSE equals its variance.
 - Squared error emphasizes large deviations, so a smaller MSE implies more consistent accuracy.
- However, Rao (1980) argued that MSE does not indicate how often an estimator lies close to the true parameter.
- Focusing solely on MSE encourages reducing the variance of unbiased estimators, hence the origin of solving for the unique minimum variance unbiased estimators (UMVUE).
- However, many useful estimators, such as the MLE, may be biased.

Estimating Alternatives to MSE

There are some alternatives to the MSE:

Mean Absolute Error

$$\mathbb{E}[|\hat{\theta} - \theta|]$$

Root Mean Squared Error

$$\sqrt{\mathbb{E}[(\hat{\theta} - \theta)^2]}$$

You do not need to memorize these.

Another possible way to compare estimators is to use the Pitman closeness (PC) criterion.

Estimating Alternatives to MSE

Pitman Closeness Criterion

Suppose we want to compare two estimators, $\hat{\theta}_1$ and $\hat{\theta}_2$, compared to a common parameter θ . Then, the PC of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is:

$$\pi_{\hat{\theta}_1, \hat{\theta}_2}(\theta) := \mathbb{P}(|\hat{\theta}_1 - \theta| \leq |\hat{\theta}_2 - \theta|).$$

If $\pi_{\hat{\theta}_1, \hat{\theta}_2}(\theta) > \frac{1}{2}$ then $\hat{\theta}_1$ is said to be **Pitman closer** to θ , implying it is a more desirable estimator.

The PC also has an intuitive appeal in applied settings. For example, Keating and Mason (1985) noted that in elections, voters typically support the candidate whose position is *probably* the closest to their own, not the one minimising squared differences. Similarly, customers choose the convenience store that *probably* has the shortest distance.

Common misconception! PC does not evaluate the magnitude of how close an estimator is to the parameter.

Estimating Alternatives to MSE

- Despite its flaws, the MSE criterion has dominated statistics education and research.
- It is also genuinely the easiest to compute.
- Others, especially the PC criterion, is much more complicated to derive exact expressions for.
- Monte Carlo methods provide a practical alternative: the PC criterion can be simulated directly rather than derived in closed form.

Estimating Alternatives to MSE

Example

Suppose $X_1, X_2, \dots, X_n$ are iid $N(\theta, \theta^2)$. We want to estimate θ . We have two possible candidates:

$$\hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{\theta}_2 = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Compute the PC probability of these two estimators and use the PC criterion to determine which one is “better”.

Solution. Consult the R codes. The second estimator should still be “better”; in this case, we say it is “Pitman closer”.

Estimating Confidence Levels

- Let $\hat{\theta}$ be a statistic that is Gaussian distributed with mean θ and standard error $\sigma_{\hat{\theta}}$.
- As $n \rightarrow \infty$, $Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}}$ has a standard Gaussian distribution as well.
- The endpoints of a $100(1 - \alpha)\%$ confidence interval for θ is given by:

$$\hat{\theta}_L = \hat{\theta} - z_{\alpha/2}\sigma_{\hat{\theta}}, \quad \hat{\theta}_U = \hat{\theta} + z_{\alpha/2}\sigma_{\hat{\theta}},$$

where $z_{\alpha/2}$ is defined as the value where $\mathbb{P}(Z > z_{\alpha/2}) = \alpha/2$ is true.

- In most cases, we won't know $\sigma_{\hat{\theta}}$ and we are generating from a sample of size n ; thus, we may use the t -distribution and $t_{\alpha/2, n-1}$ instead (where $n - 1$ represents the degree of freedom.)

Estimating Confidence Levels

To generate a confidence level from the previous scenario, employ the following:

1. Generate m random samples $x^{(1)}, x^{(2)}, \dots, x^{(m)}$ from a Gaussian distribution where $x^{(j)}$ are of size n for $j = 1, 2, \dots, m$.
2. For each replicate $x^{(j)}$:
 - 2.1 Compute the statistic $\hat{\theta}^{(j)}$
 - 2.2 Compute the sample standard error $\hat{\sigma}_{\hat{\theta}}^{(j)}$.
 - 2.3 Compute the upper and lower bounds:

$$\hat{\theta}_L^{(j)} = \hat{\theta}^{(j)} - t_{\alpha/2, n-1} \hat{\sigma}_{\hat{\theta}}^{(j)}, \quad \hat{\theta}_U^{(j)} = \hat{\theta}^{(j)} + t_{\alpha/2, n-1} \hat{\sigma}_{\hat{\theta}}^{(j)},$$

- 2.4 Compute $y_j = \mathbf{I}(\hat{\theta}_L^{(j)} \leq \theta \leq \hat{\theta}_U^{(j)})$
3. Compute the empirical confidence level $\bar{y} = \frac{1}{m} \sum_{j=1}^m y_j$

Estimating Confidence Levels

Example

Generate a Gaussian sample of size 30 with mean 5 and variance 2. Using 10000 replicates, compute the empirical confidence level for $\hat{\mu} = \sum_{i=1}^n x_i$ given $\alpha = 0.05$.

Solution. Consult the R codes.

(Aside) Rant About Statistics

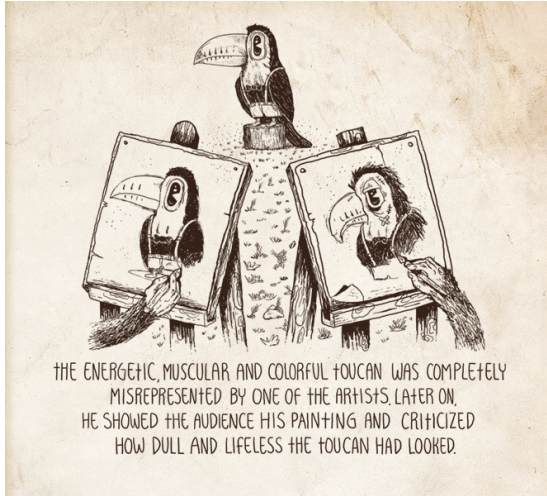
**If I could only pick one thing for everyone to learn
from this course, it would be this rant.**

I don't believe we should be doing hypothesis testing.

This take is not unique.

Hypothesis testing amounts to disproving a strawman.

What is a “Strawman?”



THE ENERGETIC, MUSCULAR AND COLORFUL TOUCAN WAS COMPLETELY MISREPRESENTED BY ONE OF THE ARTISTS. LATER ON, HE SHOWED THE AUDIENCE HIS PAINTING AND CRITICIZED HOW DULL AND LIFELESS THE TOUCAN HAD LOOKED.

From “An Illustrative Book of Bad Arguments” (highly recommended.)

Null Hypothesis Is Never True

T-Test Hypothesis Testing Set-Up

$$H_0 : \mu_1 - \mu_2 = 0, \quad H_1 : \mu_1 - \mu_2 \neq 0$$

Key questions to think about:

- Why do we say we “fail to reject” rather than “accept” the null hypothesis?
- Is the null hypothesis ever literally true in real-world data?
- Why is 0 the benchmark difference? What does “no effect” even mean?
- Suppose we estimate $\mu_1 - \mu_2 = 0.000001$. Does that make H_0 true?
- If I don't get the results I want, can't I choose a very extreme or unrealistic null to obtain statistical significance?

What is a p -value?

Pick one (or none at all):

- (i) The probability that the null hypothesis is true.
- (ii) The probability that the alternative hypothesis is true.
- (iii) One minus the probability that the alternative hypothesis is true.
- (iv) One minus the probability of replication.

Research suggests that even statistics instructors frequently struggle to articulate a correct definition (source).

Bob on [March 20, 2019 7:30 PM at 7:30 pm](#) said:

This times 1 billion. I'm sick of hearing about students 'not understanding p values'. Who is teaching them? How do you pass a stats course and not understand p values? Why don't these students fail? Lazy and incompetent lecturers, that's the problem.

What is a p -value?

p -value

Given that the null hypothesis is true, the p -value is the probability of observing a test statistic at least as extreme as the one obtained from the data.

If the null hypothesis is never true, what is the point of computing this?

What is a p -value?

David J. Marcus on **March 21, 2019 8:18 PM at 8:18 pm** said:

You can't teach p -values correctly because they don't make sense. Andrew has the right approach: Spend a little time on them to explain why you shouldn't use them. Then teach things that do make sense and work.

- Is there a point in teaching p -values?
- Educators and students don't understand it...
- The definition of a p -value is confusing and flawed...
- Some statisticians are begging to abandon the concept of statistical significance...

What Can We Do?

- Traditional statistics education heavily emphasises hypothesis testing and p -values.
- Courses for *non-statistics* majors often teach hypothesis testing as the primary tool for making research conclusions.
- Courses for *statistics* majors also dedicate significant time to hypothesis testing and p -values.
- Many applied fields, especially in the medical and social sciences, rely on hypothesis testing and p -values to draw conclusions.
- I even wrote a hypothesis testing guide for STA304 students — a decision I have since reconsidered.
- For the record: I began questioning hypothesis testing halfway through my master's degree.
- So... can we undo the damage?

Let's Take a Breath...

Before throwing out hypothesis testing entirely, let's pause and think:

- What is the ultimate goal of statistics?
- If p -values are flawed, can they still serve a useful purpose?
- Are there practical, accessible alternatives to p -values?
- Why are we so drawn to simple, black-and-white answers?

Let's Calm Down...

- Yoav Benjamini: “In some sense it offers a first line of defense against being fooled by randomness, separating signal from noise, because the models it requires are simpler than any other statistical tool needs.”
- Ben Recht: “Statistical tests constrain outcomes in participatory systems. Engineers want to push features to get promoted; data science teams insist on AB tests to ensure these features don't harm key metrics. Drug companies want to make a ton of money; clinical trials ensure drugs aren't harmful and have a chance of being beneficial. Academics want to publish as many papers as possible to get their h-index to the moon; journals insist on some NHSTs to placate editors. The purpose of statistical tests is regulation.”

What Do I Recommend?

- An analysis based on confidence intervals is actually quite reasonable.
- You have likely noticed a similarity between the confidence interval not including 0 and a p -value being less than the α threshold.

Why I Still Teach Hypothesis Testing

- My scepticism about hypothesis testing isn't new—but it's still not mainstream.
- In many professional settings, you'll still be expected to use hypothesis testing methods.
- Part of the practical value of a university degree is preparation for professional employment.
- Interviewers still ask, "What is a p -value?"
- Alternatively, even if you don't personally follow a particular religion (Christianity, Islam, Judaism, Buddhism, Hinduism, etc.), you might still learn about it because many others do, and understanding their beliefs helps you engage with them more thoughtfully.
- In the same way, even if I don't "believe in" hypothesis testing, it's important to understand it because many statisticians do.

Is This Rant Testable?

YES.

Monte Carlo Hypothesis Testing

Hypothesis Testing

Type I Error / False Positive

$$\mathbb{P}(\text{reject } H_0 | H_0 \text{ is true})$$

Type II Error / False Negative

$$\mathbb{P}(\text{fail to reject } H_0 | H_0 \text{ is false})$$

Power

$$\mathbf{Power}(\theta) = 1 - \beta = \mathbb{P}(\text{reject } H_0 | H_0 \text{ is false})$$

We can simulate these values.

Hypothesis Testing

To calculate the **type-I error rate**, we assume that the null hypothesis is true. Therefore, we simulate:

1. Generate m random samples $x^{(1)}, x^{(2)}, \dots, x^{(m)}$ from the **null** distribution where $x^{(j)}$ are of size n for $j = 1, 2, \dots, m$.
2. For each replicate $x^{(j)}$:
 - 2.1 Compute the test statistic T_j .
 - 2.2 Record the following:

$$I_j = \begin{cases} 1 & H_0 \text{ is rejected at significance level } \alpha, \\ 0 & \text{otherwise.} \end{cases}$$

3. Compute the type-I error rate by looking at the proportion of tests where H_0 was rejected: $\frac{1}{m} \sum_{j=1}^m I_j$.

Hypothesis Testing

Example

Suppose that $X_1, \dots, X_{20}$ is a random sample from a $N(\mu, \sigma^2 = 100^2)$ distribution. Test $H_0 : \mu = 500$, $H_1 : \mu \neq 500$ at $\alpha = 0.05$ where our estimator of μ is $\bar{X}$. Compute type-I error.

Solution. Under the null hypothesis we can use the following:

$$Z^* = \frac{\bar{X} - 500}{100/\sqrt{20}}$$

Consult the rest of the R codes.

Hypothesis Testing

We can simulate the **type-II error rate** and **power** if we are assuming a simple vs. simple test, i.e., the alternative hypothesis is well defined. Assume that the null hypothesis is false. Therefore, we simulate:

1. Generate m random samples $x^{(1)}, x^{(2)}, \dots, x^{(m)}$ from the **alternative** distribution where $x^{(j)}$ are of size n for $j = 1, 2, \dots, m$.
2. For each replicate $x^{(j)}$:
 - 2.1 Compute the test statistic T_j .
 - 2.2 Record the following:

$$I_j = \begin{cases} 0 & H_0 \text{ is rejected at significance level } \alpha, \\ 1 & \text{otherwise.} \end{cases}$$

3. Compute the type-II error rate by looking at the proportion of tests where H_0 was not rejected: $\frac{1}{m} \sum_{j=1}^m I_j$.

Remember: **Power** $(\theta) = 1 - \beta$.

Hypothesis Testing

Example

Using the same example as the previous example, assume $H_1 : \mu = 600$. Now compute type-I error and power. Also, test for different values for the alternative hypothesis to see how we can inflate our power!!

Solution. Consult the R codes.

Hypothesis Testing

More Formal/General) Hypothesis Testing

Let θ denote a population parameter and let Θ represent the parameter space, i.e., $\theta \in \Theta$. We say the null hypothesis is $H_0 : \theta \in \Theta_0$ and the alternative hypothesis is $H_1 : \theta \in \Theta_0^c$.

α

Let $\pi(\theta)$ denote the probability of rejecting H_0 . Then,

$$\alpha = \sup_{\theta \in \Theta_0} \pi(\theta).$$

Likelihood Ratio Test (LRT)

Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ represent a vector of realisations of the data. The likelihood ratio test *statistic* for testing $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \in \Theta_0^c$ is:

$$\lambda(\mathbf{x}) = \frac{\sup_{\Theta_0} L(\theta|\mathbf{x})}{\sup_{\Theta} L(\theta|\mathbf{x})}$$

A *likelihood ratio test* is any test that has a rejection region of the form $\{\mathbf{x} : \lambda(\mathbf{x}) \leq c\}$, where c is any number satisfying $0 \leq c \leq 1$.

Aside: Supremum/Infimum

Let S represent a subset of a partially ordered set P .

Supremum (Least Upper Bound)

An upper bound b of S is called a supremum of S if for all upper bounds z of S in P , $z \geq b$.

Infimum (Greatest Lower Bound)

A lower bound a of S is called an infimum of S if for all lower bounds y of S in P , $y \leq a$.

These concepts are similar to max/mins but sometimes the max or min doesn't exist. Consider the following:

Example

For $f(x) = e^x$ the infimum is 0 and for $g(x) = -e^x$ the supremum is 0.

Hypothesis Testing

- Earlier, my main criticisms of hypothesis testing were that you're arguing against a strawman. Most methods we use assume a simple null hypothesis.
- However, my criticism is less valid if we instead assume an interval... (However, most people DO NOT construct a hypothesis test by hand.) I'm going to show how we can do a more reasonable hypothesis test.

Simulating a More Complicated Null Hypothesis

Suppose that we had $n = 30$ patients who were sick and we give them some **drugs**. We assume that the amount of time it takes for them to recover in days is Gaussian distributed with $H_0 : \mu \geq 7$. Hence, our alternative hypothesis is $H_1 : \mu < 7$ (that is, if they take less than 7 days to recover, we consider the medicine to be effective.)

Hypothesis Testing

- Following the LRT method, we have that $L(\theta|\mathbf{x})$ is maximized at $\bar{x}$. However, under the null hypothesis we restrict $\mu \geq 7$ and thus consider:

$$\mu_{null} = \begin{cases} \bar{x} & \bar{x} \geq 7, \\ 7 & \bar{x} < 7. \end{cases}$$

- Thus,

$$\lambda(x) = \frac{\sup_{\mu \geq 7} L(\theta|\mathbf{x})}{\sup_{\mu \in \mathbb{R}} L(\theta|\mathbf{x})} = \begin{cases} 1 & \bar{x} \geq 7, \\ \frac{(2\pi\sigma^2)^{-n/2} \exp\left\{\frac{-1}{2\sigma^2} \sum_{i=1}^n (x_i - 7)^2\right\}}{(2\pi\sigma^2)^{-n/2} \exp\left\{\frac{-1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2\right\}} & \bar{x} < 7. \end{cases}$$

- Note that if $\lambda(x) = 1$ then no sample in the region ever causes rejection, so the Type-I error comes from the region $\{\bar{x} < 7\}$.

Hypothesis Testing

Simplifying the last part further,

$$\begin{aligned}\frac{\exp \left\{ \frac{-1}{2\sigma^2} \sum_{i=1}^n (x_i - 7)^2 \right\}}{\exp \left\{ \frac{-1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 \right\}} &= \exp \left\{ \frac{-1}{2\sigma^2} \sum_{i=1}^n [(x_i - 7)^2 - (x_i - \bar{x})^2] \right\} \\ &= \exp \left\{ \frac{-1}{2\sigma^2} n(\bar{x} - 7)^2 \right\} \quad (\text{Details are left as an exercise!})\end{aligned}$$

Thus the rejection region (RR) is (again, details are left as an exercise):

$$\left\{ \mathbf{x} : \exp \left\{ \frac{-1}{2\sigma^2} n(\bar{x} - 7)^2 \right\} \leq c \right\} = \left\{ \mathbf{x} : \left(\frac{\bar{x} - 7}{\sigma/\sqrt{n}} \right)^2 > c' \right\}$$

Where $\left(\frac{\bar{x} - 7}{\sigma/\sqrt{n}} \right)^2 \sim \chi_{(1)}^2$. If we set $\alpha = 0.05$, then $c' = \chi_{df=1, \alpha=0.05}^2 = 3.841459$.

Hypothesis Testing

Disclaimer! This is NOT a Monte Carlo simulation!

```
n <- 30; sig <- 2.5 # assuming variance is known...
# This is simulated but a placeholder to put real data
dat <- rnorm(30, mean = 4, sd = sig)
if(mean(dat) >= 7){
  print("fail to reject")
} else{
  c <- qchisq(0.05, df = 1, lower.tail = FALSE)
  crit.region <- ((mean(dat)-7)/(sig/sqrt(n)))^2
  if(crit.region > c){
    print("reject")
  } else {
    print("fail to reject")
  }
}
```